

Mallika Laasya Ramachandrani

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EDUCATION

University of Wisconsin – Madison

B.S. Computer Science & Economics

Madison, WI

Expected May 2028

- Relevant Coursework: Matrix methods in Machine Learning, Programming 3, Linear Algebra, Statistics in Economics

EXPERIENCE

Undergraduate Learning Center – CS Tutor

Aug. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- Tutor 100s of students across CS 200/300 in Java fundamentals, OOP inheritance, recursion, ArrayLists, HashMaps, and Big-O time complexity analysis
- Engineered supplemental problem sets and worked through targeted debugging sessions, contributing to improvement in section quiz scores
- Conduct weekly group office hours and one-on-one sessions, leading to higher student satisfaction

Pain Context Innovation Lab – Research Assistant

Jan. 2026 – Present

University of Wisconsin – Madison | PI: Dr. Shinye Kim

Madison, WI

- Architected and deployed the clinic navigation module for a patient-facing web platform serving 100s of chronic pain patients, reducing average time-to-clinic-match by ~40%
- Integrated NPPES NPI REST API, dynamically querying and filtering 1M+ clinician records by insurance type, specialty, and geographic location to surface relevant providers
- Developed responsive search and filter UI components in React with real-time state management, cutting average user lookup time from 5+ min to under 60 sec in internal testing
- Collaborated with a 5-person cross-functional team of clinicians and developers across bi-weekly sprints, managing feature branches and conducting pull request reviews on GitHub

PROJECTS

VitalLens | [GitHub](#) | React, TypeScript, Supabase, Claude API, Recharts

April 2026

- Built a full-stack AI health companion for 100s of users; engineered a PDF parsing pipeline extracting and classifying 20+ biomarker types with structured JSON output via Claude Sonnet 4.6
- Designed a multi-visit longitudinal dashboard using Recharts to visualize biomarker trends over time; integrated automated doctor prep question generation, reducing patient appointment prep time from 30+ min to under 5 min
- Implemented Supabase Auth with Row-Level Security policies and PostgreSQL Edge Functions, ensuring HIPAA-aligned per-user data isolation with zero cross-user data exposure

Vocalis | [GitHub](#) | Next.js, TypeScript, FastAPI, Groq, Whisper, Docker, Firebase

June 2026

- Developed a full-stack AI mock interview platform for 100s of students, generating role-specific technical and behavioral questions across 3 engineering disciplines via Groq LLMs and a RESTful FastAPI backend
- Built end-to-end voice pipeline integrating OpenAI Whisper with the Web Audio API for real-time audio capture and transcription, achieving <2s latency on responses up to 3 minutes in length
- Engineered a scikit-learn answer evaluation model scoring responses across 4 dimensions (clarity, relevance, depth, communication) with structured JSON feedback delivered to the frontend in under 3 seconds
- Containerized the entire backend stack (FastAPI + SQLAlchemy + SQLite) using Docker Compose and deployed with a Firebase-authenticated CI/CD pipeline, enabling reproducible one-command local setup

CAMPUS INVOLVEMENT

Women in Business Technology – VP of Systems

Jan. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- Elected VP of Systems for Fall 2026; architect and maintain club website for 100+ member organization, shipping 10+ feature updates per semester and integrating professional networking resources

WACM – Undergraduate Activities Chair

Feb. 2026 – Present

University of Wisconsin – Madison

Madison, WI

- Selected as Undergraduate Activities Chair for Fall 2026; curate and organize monthly technical speaker series for 50+ members featuring engineers from Google, Microsoft, and Amazon

AWARDS & CERTIFICATIONS

- 2nd Place – BadgerAI Hackathon (80+ participants), April 2026
- 3rd Place – Women in Business × Uline Case Competition, UW–Madison, October 2025

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, HTML/CSS, SQL | **Frameworks:** React, Next.js, FastAPI, Node.js, Tailwind CSS, scikit-learn

Tools: Git, Docker, Firebase, Supabase, GitHub Actions, Groq API, Claude API, Whisper, VS Code